



Alberta Public Health Disease Management Guidelines

Smallpox



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Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

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Case Definition

Confirmed Case

Laboratory confirmation of infection:

- Isolation of variola virus from an appropriate clinical specimen (e.g., whole blood, lesion fluid, crust material)^(A) (tier 3 laboratory only)

OR

- Detection of variola virus nucleic acid (e.g., PCR) in an appropriate clinical specimen (e.g., whole blood, lesion fluid, crust material)⁽¹⁾

Probable Case

Clinical illness^(B) in a person who has an epidemiological link to a laboratory-confirmed case or to a probable case of smallpox

OR

Laboratory evidence of infection:

- Negative stain electron microscopy identification of variola virus in an appropriate clinical specimen

Suspect Case

Clinical illness^(A) in a person who does not have an epidemiological link to a laboratory-confirmed case or to a probable case of smallpox

OR

Atypical lesion^(C) known to be associated with the variola virus on a person who has an epidemiological link to a laboratory confirmed or probable case

^(A) Refer to the [National Microbiology Laboratory Guide to Services](#) for current specimen collection and submission information.

^(B) Clinical illness: Smallpox is characterized by a febrile prodrome consisting of fever > 38.3° C and systemic symptoms (prostration, headache, back pain, abdominal pain and/or vomiting) which generally lasts one to four days and is followed by the development of a characteristic rash. The rash consists of deep, firm, well-circumscribed pustules that are mostly all in the same stage of development. The lesions are characteristically umbilicated. The lesions initially appear as macules, evolving into papules, vesicles and then pustules in a matter of days. Finally, crusted scabs form; they then fall off several weeks after the initial appearance of the rash. Lesions initially appear in the oral mucosa/palate and then progress in a centrifugal pattern to involve the face, arms, legs, palms and soles. Atypical presentations include flat velvety lesions that do not evolve into pustules, and more severe forms with confluent or hemorrhagic lesions.

^(C) Atypical presentations of smallpox include: haemorrhagic lesions OR flat velvety lesions not appearing as typical vesicles nor progressing to pustules.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed, probable and suspect cases in the prescribed form by the Fastest Means Possible (FMP).

Laboratories

All laboratories shall report all positive laboratory results by FMP to the MOH (or designate) of the zone and the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall notify the CMOH (or designate) by FMP of all confirmed, probable and suspect cases, including suspected deliberate release.
- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed, probable and suspect cases to the CMOH (or designate) within one week of notification and the final NDR (amendment) within two weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by FMP:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
October 2021	General	• Updated Template • Checked web link